A Literature-Implied Partial Answer to Dai's Mixed Sequence-Space Stability Problem

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Status. `literature_implied_answer (partial subcase)`.

Source question

Dai asks in Problem 4.7(9) of arXiv:1402.2123v4, PDF page 11, whether the self-pair

$$E_{p,q} := \left(\bigoplus_{n=1}^{\infty} \ell_p\right)_{\ell_q} = \ell_q(\ell_p), \qquad (E_{p,q}, E_{p,q}),$$

is stable for standard ε-isometries [1]. Here stability means that for every standard ε-isometry $f : E_{p,q} \to E_{p,q}$ there is a bounded linear $T : E_{p,q} \to E_{p,q}$ and a constant γ such that $\|Tf(x) - x\| \le \gamma\varepsilon$ for all x.

Literature-implied subcase

Proposition 1. *Work over the real scalars. Suppose*

$$1 < p, q < \infty, \qquad p \ne q, \qquad p, q \ne 2, \qquad \text{and} \qquad \neg(1 < q < p < 2).$$

Then $(E_{p,q}, E_{p,q})$ is $(1,2)$-stable. More precisely, for every standard ε-isometry $f : E_{p,q} \to E_{p,q}$ there is a linear operator $T : E_{p,q} \to E_{p,q}$ satisfying

$$\|T\| = 1, \qquad \|Tf(x) - x\| \le 2\varepsilon \quad (x \in E_{p,q}).$$

Proof. Put $E = E_{p,q}$. Since $1 < p, q < \infty$, the real Banach space $E = \ell_q(\ell_p)$ is uniformly convex and uniformly smooth.

We first identify a consequence of two isometry/projection results. Realize the two sequence spaces as $L_q(\mathbb{N}, \mu)$ and $L_p(\mathbb{N}, \nu)$ for strictly positive finite atomic measures μ, ν; diagonal coordinate rescaling gives an isometry with the usual counting measure model. Koldobsky's Theorem 3 applies to every real linear isometry $U : E \to E$ provided, in his notation, the outer exponent $r = q$ and inner exponent $s = p$ satisfy

$$r \ne s, \quad s \ne 2, \quad s \notin (r, 2) \text{ if } r < 2.$$

The last condition is precisely the exclusion of $1 < q < p < 2$. The theorem writes U in weighted-rearrangement/fibre form [2, Theorem 3, pp. 697–699].

Let (e_{ij}) be the canonical double basis of E, with i the outer coordinate, and write $y_{ij} = Ue_{ij}$. The atomic form of Koldobsky's description has the following consequences. For fixed i, all y_{ij} have the same outer support A_i; the sets A_i are disjoint as i varies. At every $k \in A_i$,

$$y_{ij}(k) = h_i(k)V_{i,k}e_j,$$

where $V_{i,k} : \ell_p \to \ell_p$ is a linear isometry. Since $p \neq 2$, the classical Lamperti description of into isometries of ℓ_p implies that $(V_{i,k}e_j)_j$ is a family of disjointly supported unit vectors. Consequently, for distinct double indices, either the outer supports are disjoint, or they are equal and the corresponding inner vectors are disjoint with equal ℓ_p-norm at each outer coordinate.

Apply Theorem 5.1 of Lemmens–Randrianantoanina–van Gaans with outer exponent q and inner exponent p. Its hypotheses also require $p, q \neq 2$, and the basis just described satisfies its exact criterion. It follows that $U(E)$ is the range of a projection $P : E \to U(E)$ with $\|P\| = 1$ [3, Theorem 5.1, p. 19].

Now let $f : E \to E$ be a standard ε-isometry. Uniform convexity gives the real linear isometry

$$\varphi(x) = \lim_{t\to\infty} \frac{f(tx)}{t}.$$

By the preceding paragraph there is a norm-one projection $P : E \to \varphi(E)$. Remarks 2.5(2) of Šemrl–Väisälä state that their proof applies to any uniformly convex and uniformly smooth codomain for which this asymptotic copy is the range of a norm-one projection. With $T = \varphi^{-1}P$, that proof gives

$$\|T\| = 1, \qquad \|Tf(x) - x\| \leq 2\varepsilon \quad (x \in E)$$

[4, Remarks 2.5(2), p. 9]. This is the asserted $(1, 2)$-stability. □

Provenance, search, and limitations

The supporting results date from 1991, 2002, and 2005/2007, before the source question was posted. Their relation to Problem 4.7(9) is an identification made in this run; none of the supporting papers says that it answers Dai's later question. Exact-question and close-variant searches did not locate a later paper explicitly advertising this subcase as a solution. The result is therefore recorded as a literature-implied partial answer, not as a new theorem.

The proof does not cover $p = 1$, $q = 1$, either exponent equal to 2, the remaining real region $1 < q < p < 2$, any infinite endpoint, the L_p-fibre problem in item (10), or the other items of Problem 4.7.

References

[1] D. Dai, *On Qian's problem for $\mathcal{L}_\infty$-spaces*, arXiv:1402.2123v4, 2018.

[2] A. L. Koldobsky, *Isometries of $L_p(X; L_q)$ and equimeasurability*, Indiana Univ. Math. J. **40** (1991), no. 2, 677–705, doi:10.1512/iumj.1991.40.40031.

[3] B. Lemmens, B. Randrianantoanina, and O. van Gaans, *Second derivatives of norms and contractive complementation in vector-valued spaces*, arXiv:math/0511044, 2005.

[4] P. Šemrl and J. Väisälä, *Nonsurjective nearisometries of Banach spaces*, arXiv:math/0112294v2, 2002.